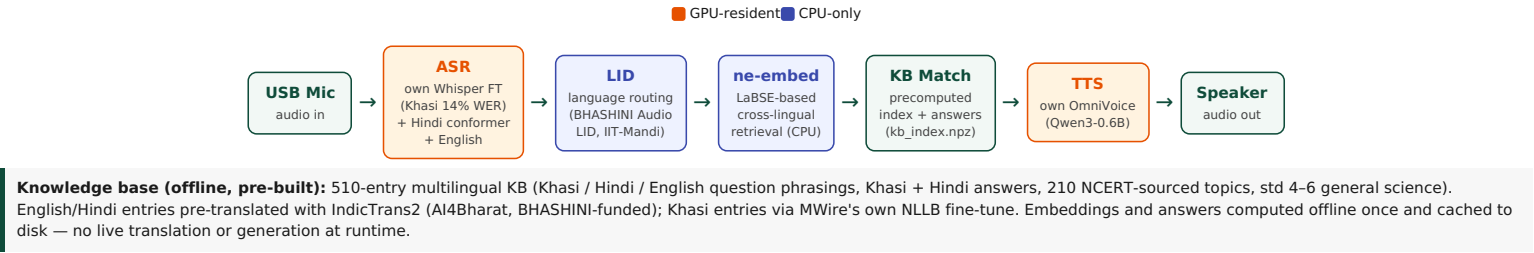


VYOMA Innovation Challenge — Technical Architecture

MWire Labs — Suno Sutra Prototype: Khasi Education Assistant (offline voice Q&A, std 4–6 general science)

1. System Architecture



2. Key Design Choices

- Retrieval-first, no generative LLM:** ASR → embedding-based KB match → TTS. Removes hallucination risk and cuts compute, at the cost of scope being limited to curated content.
- Own NE-language stack:** Khasi ASR, MT, TTS, and embedding models are MWire's own fine-tunes, purpose-built for low-resource Northeast Indian languages rather than adapted general-purpose models.
- Offline-first:** all models and the KB are resident on-device; no live API calls at inference time.
- Pre-translated KB over live MT:** translation moved offline into a one-time KB-build step; at runtime only cross-lingual embedding match is needed, removing an entire model from the live GPU path.
- Multilingual by design:** Khasi (primary), Hindi, and English all route through the same LID → retrieval → TTS pipeline, demonstrated live during the demo by the presenter.
- Interaction:** touchscreen is the primary ask/interact trigger; the physical capacitive touch sensor is repurposed as a no-look "repeat last answer" control.

3. Engineering Quality & Reliability

GPU memory constraint solved: the Jetson Orin Nano's default CMA (contiguous memory allocator) pool is only 256MB, separate from general RAM. Loading 3+ transformer models GPU-resident simultaneously (ASR + MT + TTS) caused a hardware-level NvMap/CMA crash. Rather than risk the CUDA driver by editing boot config, the pipeline was re-architected: MT was moved offline into the KB-build step, so only ASR and TTS remain GPU-resident at runtime; LID and retrieval (ne-embed) run CPU-only. This was validated end-to-end and is the pipeline shipped in the prototype.

- Deployment host: Jetson Orin Nano 8GB ("pocket-infer-0935"), JetPack 6.2.1, SSH-managed.
- Model provenance: Badnyal/kha-en-nllb-v0.4 (MT, KB-build only), MWirelabs/NE-TTS-kha-voice-01 (TTS), MWirelabs/ne-asr (ASR), MWirelabs/ne-embed (retrieval) — all MWire Labs models on Hugging Face.
- Hindi handled via a dedicated conformer + CTC decoder path bundled in the ASR repo (accurate Devanagari output), not via the Whisper model's own Hindi token, which degrades under the NE-focused fine-tune.

4. Latency

Stage	Compute	Notes
ASR	GPU	Khasi / Hindi (dedicated path) / English
LID + retrieval (ne-embed)	CPU	runs alongside GPU stages, not a bottleneck
TTS	GPU	OmniVoice, Qwen3-0.6B backbone
Warm end-to-end (validated)	—	~2.1-2.4s, against a target budget of 5-8s

5. Accuracy

- Khasi ASR: ~14% WER on own fine-tuned Whisper model.
- Khasi↔English MT (used offline for KB translation, not live inference): BLEU ~27 (kha→en) / ~25 (en→kha).
- BHASHINI-linked components: BHASHINI Audio LID (IIT-Mandi, 12 languages) for language routing, and IndicTrans2 (AI4Bharat, BHASHINI-funded) for offline KB translation — satisfying the challenge's BHASHINI-inclusion requirement.

6. Power Efficiency

- Single Jetson Orin Nano 8GB module handles the full pipeline; no external server or cloud GPU dependency.
- Device is mains-powered/stationary for the prototype (kit built for plugged operation, not battery).
- CPU-only LID and retrieval stages reduce simultaneous GPU load, keeping thermal/power draw lower than an always-GPU-resident pipeline.

7. How This Advances Open, Multilingual, Inclusive, Privacy-Preserving, Efficient AI

- **Open & multilingual:** purpose-built ASR/MT/TTS/embedding models for Khasi — a language with zero to partial BHASHINI coverage — built on and interoperating with BHASHINI's open ecosystem (IndicTrans2, Audio LID).
- **Inclusive:** targets std 4-6 learners in a language most large AI systems don't support, with a stationary offline device removing the need for internet access or a smartphone.
- **Privacy-preserving:** fully offline inference — no audio, query, or usage data leaves the device.
- **Efficient:** single edge-compute board, no generative LLM, retrieval-first design keeps latency and power draw low.